higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

FLUID MECHANICS N6

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This marking guideline consists of 7 pages.

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QUESTION 1

1.1 A flow in which particles move in an orderly manner and retain their relative positions in successive cross sections. (2)

1.2 Portion of pipe in contact with fluid (1)

1.3 1.3.1 $Q = \frac{820}{1000 \text{ x } 60}$ ✓
$= 0,014 \text{ m}^3/\text{s}$ ✓

1.3.2 $S = \pi DL$
$= \pi \times 0,36 \times 45$ ✓
$= 50,894 \text{ m}^2$ ✓

1.3.3 $A = \frac{\pi D^2}{4}$
$= \frac{\pi (0,36)^2}{4}$ ✓
$= 0,102 \text{ m}^2$ ✓

1.3.4 $h = 45 \sin 35^\circ$ ✓
$= 25,811 \text{ m}$ ✓

1.3.5 $V = \frac{Q}{A}$
$= \frac{0,014}{0,102}$ ✓
$= 0,134 \text{ m/s}$ ✓

(5 × 2) (10)

1.4 $V_1 = \frac{0,08 \text{ x } 4}{\pi \text{ x } (0,35)^2} = 0,832 \text{ m/s}$ ✓
$V_2 = \frac{0,08 \text{ x } 4}{\pi \text{ x } (0,15)^2} = 4,527 \text{ m/s}$ ✓

$Q = \frac{288}{3600} = 0,08 \text{ m}^3/\text{s}$ ✓
$\frac{Pr_1}{\rho g} + \frac{v^2}{2g} + z_1 = \frac{Pr_2}{\rho g} + \frac{v^2}{2g} + z_2$
$\frac{Pr_1}{9,81} + \frac{0,832^2}{2 \text{ x } 9,81} + 0 = \frac{P_{r2}}{9,81} + \frac{4,527^2}{2 \text{ x } 9,81} + 0$ ✓
$Pr_1 - Pr_2 = 19,803 \text{ kPa}$ ✓

alternatively

$Pr_2 - Pr_1 = -19,803 \text{ kPa}$ ✓ (7)

1.5 $C = \sqrt{\frac{2g}{f}}$

$= \sqrt{\frac{2\ x\ 9,81}{0,02}}$ ✓

$= 31,321$ ✓

$m = \frac{d}{4}$

$= \frac{0,4}{4}$

$= 0,1$ m ✓

$Q = \frac{2,37}{60}$

$= 0,0395\ m^3/s$ ✓

$A = \frac{\pi d^2}{4}$

$= \frac{\pi\ x\ 0,4^2}{4}$

$= 0,126\ m^2$ ✓

$Q = AC\sqrt{mi}$

$0,0395 = 0,126 \times 31,321 \sqrt{0,1\ x\ i}$ ✓

$i = 0,001$ ✓

$i = \frac{h_f}{L}$

$0,001 = \frac{h_f}{1500}$ ✓ $h_f = 1,512$ m ✓ (9)

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QUESTION 2

2.1 $A = ½ \times B \times H$

$= ½ \times 4 \times 3$ ✓

$= 6\ m^2$ ✓

$P = 2\left(\sqrt{2^2+3^2}\right)$ ✓

$= 7,211$ m ✓

$m = \frac{A}{P}$

$= \frac{6}{7,211}$ ✓

$= 0,832$ m ✓

$i = \frac{1}{2550}$

$= 0,392 \times 10^{-3}$ ✓

$Q = AC\sqrt{mi}$

$= 6 \times 50 \times \sqrt{0,832\ x\ 0,392\ x\ 10^{-3}}$ ✓

$= 5,419\ m^3/s$ (9)

= 5419,908 ℓ/s✓

2.2 $Q = Cd \times \frac{8}{15} \times \sqrt{2g} \tan\frac{\theta}{2} \times H^{2,5}$ ✓

$= 0,8 \times \frac{8}{15} \times \sqrt{2\ x\ 9,81} \tan\frac{90}{2} \times 0,895^{2,5}$ ✓

= 1,432 m³/s✓ (3)

2.3 2.3.1 $V_A = \sqrt{\frac{g x^2}{2y}}$

$= \sqrt{\frac{9,81\ x\ 1,87^2}{2(0,77)}}$ ✓

= 4,72 m/s✓

Reaction of the jet = ρQV$_A$✓

= $10^3 \times 0,24 \times 4,72$✓

= 1132,732 N✓ (5)

2.3.2 $V_A = C_V\sqrt{2gh}$

$4,72 = 0,98\sqrt{2\ x\ 9,81\ x\ h}$✓

h = 1,182 m✓ (2)

2.3.3 $Q_{th} = V_{th} \times A_{th}$

$= \sqrt{2gh} \times \frac{\pi d^2}{4}$

$= \sqrt{2\ x\ 9,81\ x\ 1,182}$✓ $\times \frac{\pi(0,05)^2}{4}$✓ (3)

= 9,456 ℓ/s✓ **[22]**

QUESTION 3

3.1
- Axial flow fans
- Centrifugal fans (2)

3.2 $a = \frac{\pi(0,75)^2}{4}$ ✓

= 0,442 m²✓

S = πDL

= π × 0,75 × 60✓

= 141,372 m²✓

$Pr = \frac{kSV^2}{a}$

$180 = \frac{0,005\ x\ 141,372\ x\ Q^2}{0,442^3}$✓

= 4,686 m³/s✓ (6)

3.3 3.3.1 $Q = \frac{42300}{1000\ x\ 3600}$✓ (4)

= 0,01175 m^3/s✓

$$P = \frac{\rho g Q H}{\eta}$$

$$= \frac{10^3 \; x \; 9,81 \; x \; 0,01175 \; x \; 258,5}{0,85}$$ ✓

= 35,055 kW✓

3.3.2 $Hf_1 = 258,5 - 206$

= 52,5 m✓

$$\frac{hf_2}{hf_1} = \left(\frac{Q_2}{Q_1}\right)^2$$

$$= \left(\frac{0,0235}{0,01175}\right)^2 \times 52,5$$ ✓

= 210 m✓

H = 210 + 206 = 416 m ✓

$$P = \frac{\rho g Q H}{\eta}$$

$$= \frac{10^3 \; x \; 9,81 \; x \; 0,0235 \; x \; 416}{0,85}$$ ✓

= 112,827 kW✓ (6)

3.4
- Reciprocating pump uses impeller to suck and deliver fluid
- Centrifugal fan uses piston/plunger to suck and deliver fluid (2)

3.5 3.5.1 Velocity of water relative to impeller

3.5.2 Velocity of water relative to pump casing

(2 × 1) (2)

3.6

$$Q = \frac{12,24 \; x \; 10^6}{24 \; x \; 10^3 \; x \; 3600}$$ ✓

= 0,142 m^3/s✓

$$h_f = \frac{fl Q^2}{3 d^5}$$

$$= \frac{0,007 \; x \; 26000 \; x \; 0,142^2}{3 \; x \; 0,45^5}$$ ✓

= 65,982 m ✓

$$P = \frac{\rho g Q H}{\eta}$$ (6)

$= \frac{10^3 \; x \; 9,81 \; x \; 0,142 \; x \; 65,982}{0,78}$ ✓

= 117,561 Kw✓

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QUESTION 4

4.1. 4.1.1

Sin 60° = $\frac{V_{wi}}{36}$ ✓

V_{wi} = 31,177 m/s✓

Alternatively

Cos30° = $\frac{V_{wi}}{36}$ ✓

V_{wi} = 31,177 m/s✓

Cos60° = $\frac{V_{wo}}{3}$ ✓

V_{wo} = 1,5 m/s✓

Alternatively

Sin30° = $\frac{V_{wo}}{3}$ ✓

V_{wo} = 1,5 m/s✓ (4)

4.1.2

$U_i = \frac{\pi \; D \; N}{60}$

$= \frac{\pi \; x \; 1,2 \; x \; 300}{60}$ ✓

= 18,849 m/s✓

(8)

$\text{Cos}60^0 = \frac{V_i}{36}$ ✓ Alternatively $\text{Sin}30^\circ = \frac{V_i}{36}$✓

V_i = 18 m/s✓ V_i = 18 m/s✓

x = V_{wi} - U_i
= 31,177 – 18,849✓
= 12,327 m/s✓

$\text{Tan}\ \beta_i = \frac{V_i}{x}$

$\beta_i = \text{Tan}^{-1}\frac{18}{12,327}$ ✓
= 55,595°✓

4.2 4.2.1 $V_A = C_V\sqrt{2gh}$

$= \sqrt{2\ x\ 9{,}81\ x\ 245}$ ✓

= 69,332 m/s✓

For maximum efficiency U = 0,5 V_A

U = 0,5 (69,332)✓

= 34,666 m/s✓

$U = \frac{\pi\ D\ N}{60}$

$34{,}666 = \frac{\pi\ x\ 0{,}9\ x\ N}{60}$ ✓

= 735,633 r/min✓ (6)

4.2.2 $\eta = \frac{U}{gh}$ (V – U) (1 + cos (180° – y) × 100%✓

$= \frac{34{,}666}{9{,}81\ x\ 245}$ (69,332 – 34,666) [1 + cos (180° – 160°)] × 100%✓ (3)

= 96,985%✓ **[21]**

TOTAL: 100

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